



2009
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{\ln x}{x} dx$. **2**

(b) Find $\int x e^{2x} dx$. **2**

(c) Find $\int \frac{x^2}{1 + 4x^2} dx$. **3**

(d) Evaluate $\int_2^5 \frac{x - 6}{x^2 + 3x - 4} dx$. **4**

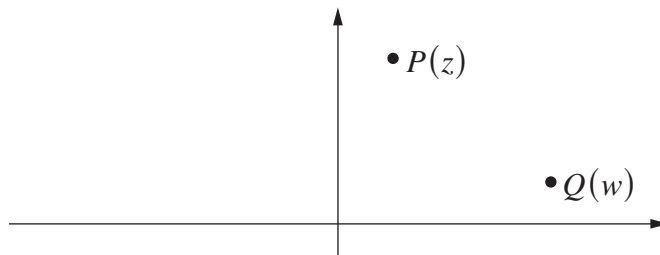
(e) Evaluate $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1 + x^2}} dx$. **4**

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) Write i^9 in the form $a + ib$ where a and b are real. **1**

(b) Write $\frac{-2 + 3i}{2 + i}$ in the form $a + ib$ where a and b are real. **1**

(c) The points P and Q on the Argand diagram represent the complex numbers z and w respectively.



Copy the diagram into your writing booklet, and mark on it the following points:

(i) the point R representing iz **1**

(ii) the point S representing $\bar{w}$ **1**

(iii) the point T representing $z + w$. **1**

(d) Sketch the region in the complex plane where the inequalities $|z - 1| \leq 2$ and $-\frac{\pi}{4} \leq \arg(z - 1) \leq \frac{\pi}{4}$ hold simultaneously. **2**

(e) (i) Find all the 5th roots of -1 in modulus-argument form. **2**

(ii) Sketch the 5th roots of -1 on an Argand diagram. **1**

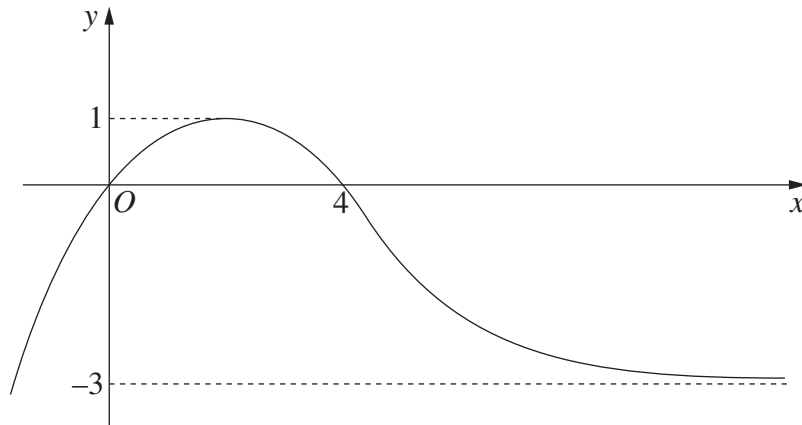
(f) (i) Find the square roots of $3 + 4i$. **3**

(ii) Hence, or otherwise, solve the equation **2**

$$z^2 + iz - 1 - i = 0.$$

Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the graph $y = f(x)$.



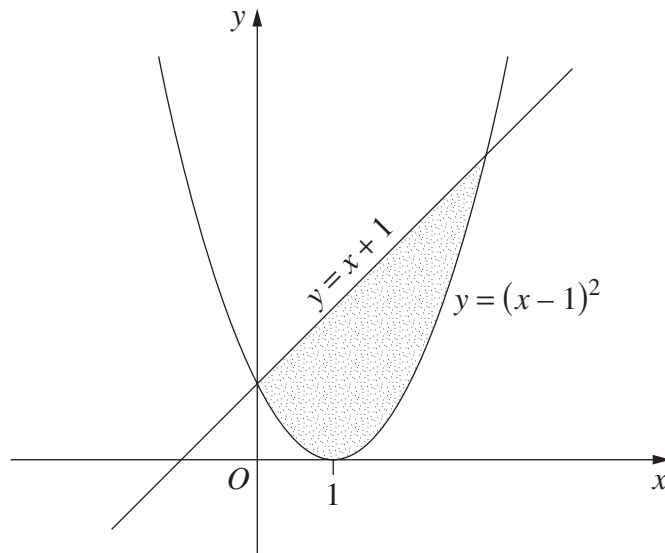
Draw separate one-third page sketches of the graphs of the following:

- (i) $y = \frac{1}{f(x)}$ **2**
- (ii) $y = [f(x)]^2$ **2**
- (iii) $y = f(x^2)$. **2**
- (b) Find the coordinates of the points where the tangent to the curve $x^2 + 2xy + 3y^2 = 18$ is horizontal. **3**
- (c) Let $P(x) = x^3 + ax^2 + bx + 5$, where a and b are real numbers. **3**
- Find the values of a and b given that $(x - 1)^2$ is a factor of $P(x)$.

Question 3 continues on page 5

Question 3 (continued)

- (d) The diagram shows the region enclosed by the curves $y = x + 1$ and $y = (x - 1)^2$. **3**



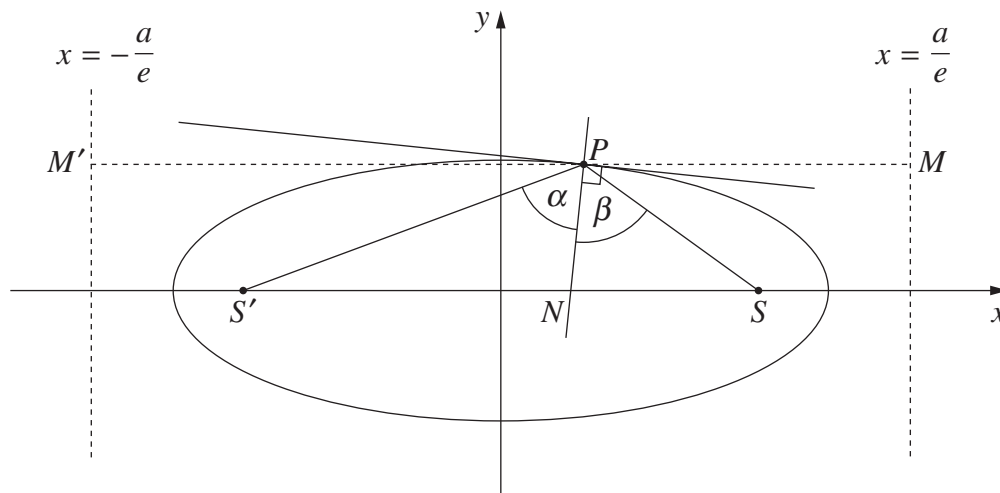
The region is rotated about the y-axis.

Use the method of cylindrical shells to find the volume of the solid formed.

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has foci $S(ae, 0)$ and $S'(-ae, 0)$ where e is the eccentricity, with corresponding directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$. The point $P(x_0, y_0)$ is on the ellipse. The points where the horizontal line through P meets the directrices are M and M' , as shown in the diagram.



- (i) Show that the equation of the normal to the ellipse at the point P is 2

$$y - y_0 = \frac{a^2 y_0}{b^2 x_0} (x - x_0).$$

- (ii) The normal at P meets the x -axis at N . Show that N has coordinates $(e^2 x_0, 0)$. 2

- (iii) Using the focus-directrix definition of an ellipse, or otherwise, show that 2

$$\frac{PS}{PS'} = \frac{NS}{NS'}.$$

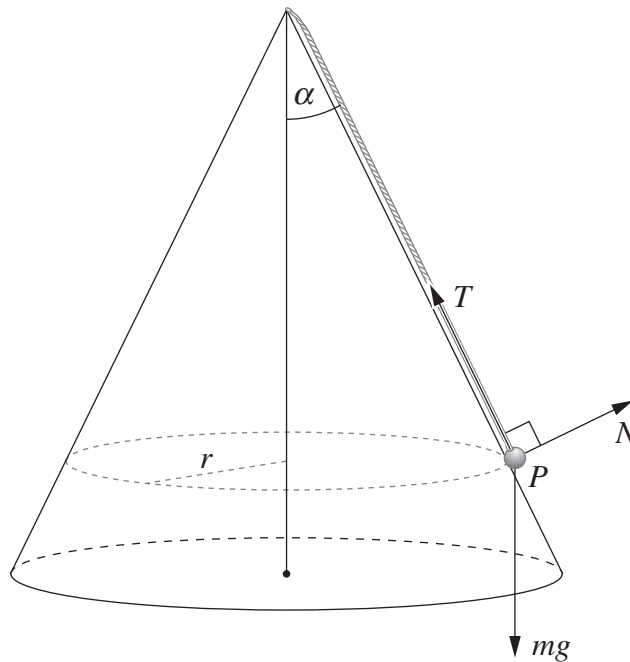
- (iv) Let $\alpha = \angle S'PN$ and $\beta = \angle NPS$. 2

By applying the sine rule to $\triangle S'PN$ and to $\triangle NPS$, show that $\alpha = \beta$.

Question 4 continues on page 7

Question 4 (continued)

- (b) A light string is attached to the vertex of a smooth vertical cone. A particle P of mass m is attached to the string as shown in the diagram. The particle remains in contact with the cone and rotates with constant angular velocity ω on a circle of radius r . The string and the surface of the cone make an angle of α with the vertical, as shown.



The forces acting on the particle are the tension, T , in the string, the normal reaction, N , to the cone and the gravitational force mg .

- (i) Resolve the forces on P in the horizontal and vertical directions. 2
- (ii) Show that $T = m(g \cos \alpha + r\omega^2 \sin \alpha)$ and find a similar expression for N . 2
- (iii) Show that if $T = N$ then 2

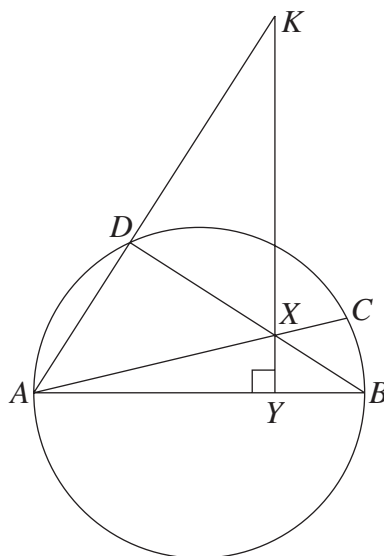
$$\omega^2 = \frac{g}{r} \left(\frac{\tan \alpha - 1}{\tan \alpha + 1} \right).$$

- (iv) For which values of α can the particle rotate so that $T = N$? 1

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) In the diagram AB is the diameter of the circle. The chords AC and BD intersect at X . The point Y lies on AB such that XY is perpendicular to AB . The point K is the intersection of AD produced and YX produced.



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) Show that $\angle AKY = \angle ABD$. | 2 |
| (ii) Show that $CKDX$ is a cyclic quadrilateral. | 2 |
| (iii) Show that B, C and K are collinear. | 2 |

Question 5 continues on page 9

Question 5 (continued)

(b) For each integer $n \geq 0$, let

$$I_n = \int_0^1 x^{2n+1} e^{x^2} dx.$$

(i) Show that for $n \geq 1$

2

$$I_n = \frac{e}{2} - nI_{n-1}.$$

(ii) Hence, or otherwise, calculate I_2 .

2

(c) Let $f(x) = \frac{e^x - e^{-x}}{2} - x$.

(i) Show that $f''(x) > 0$ for all $x > 0$.

2

(ii) Hence, or otherwise, show that $f'(x) > 0$ for all $x > 0$.

2

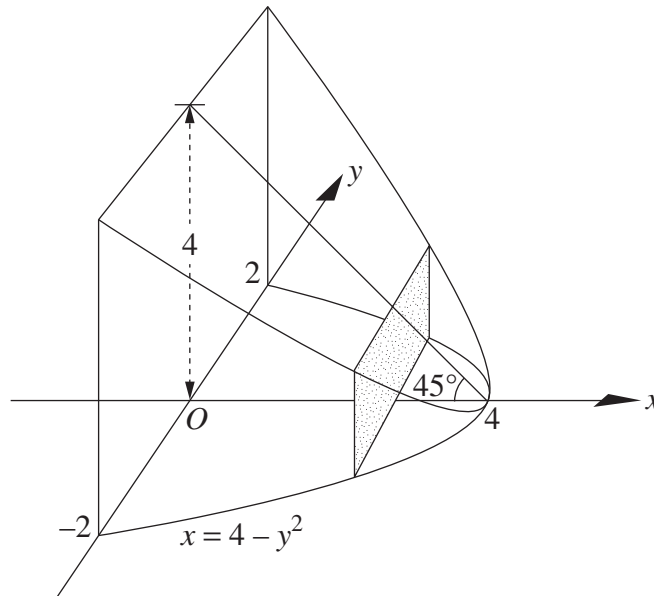
(iii) Hence, or otherwise, show that $\frac{e^x - e^{-x}}{2} > x$ for all $x > 0$.

1

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) The base of a solid is the region enclosed by the parabola $x = 4 - y^2$ and the y -axis. The top of the solid is formed by a plane inclined at 45° to the xy -plane. Each vertical cross-section of the solid parallel to the y -axis is a rectangle. A typical cross-section is shown shaded in the diagram. **3**



Find the volume of the solid.

- (b) Let $P(x) = x^3 + qx^2 + qx + 1$, where q is real. One zero of $P(x)$ is -1 .

(i) Show that if α is a zero of $P(x)$ then $\frac{1}{\alpha}$ is a zero of $P(x)$. **1**

(ii) Suppose that α is a zero of $P(x)$ and α is not real.

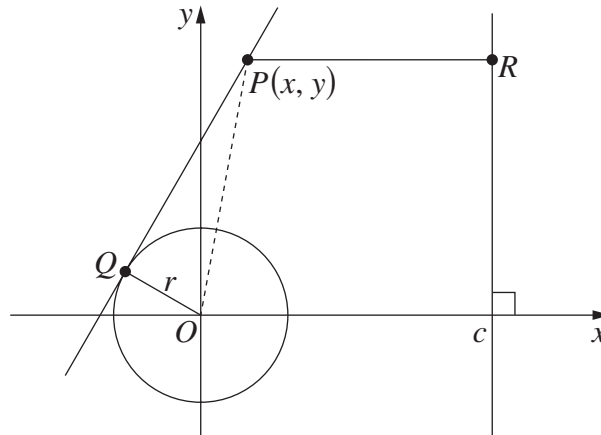
(1) Show that $|\alpha| = 1$. **2**

(2) Show that $\operatorname{Re}(\alpha) = \frac{1-q}{2}$. **2**

Question 6 continues on page 11

Question 6 (continued)

- (c) The diagram shows a circle of radius r , centred at the origin, O . The line PQ is tangent to the circle at Q , the line PR is horizontal, and R lies on the line $x = c$.

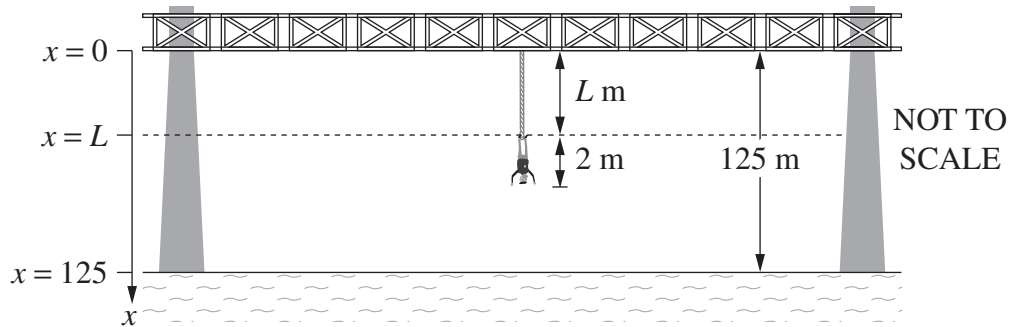


- (i) Find the length of PQ in terms of x , y and r . **1**
- (ii) The point P moves such that $PQ = PR$. **2**
- Show that the equation of the locus of P is
- $$y^2 = r^2 + c^2 - 2cx.$$
- (iii) Find the focus, S , of the parabola in part (ii). **2**
- (iv) Show that the difference between the length PS and the length PQ is independent of x . **2**

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) A bungee jumper of height 2 m falls from a bridge which is 125 m above the surface of the water, as shown in the diagram. The jumper's feet are tied to an elastic cord of length L m. The displacement of the jumper's feet, measured downwards from the bridge, is x m.



The jumper's fall can be examined in two stages. In the first stage of the fall, where $0 \leq x \leq L$, the jumper falls a distance of L m subject to air resistance, and the cord does not provide resistance to the motion. In the second stage of the fall, where $x > L$, the cord stretches and provides additional resistance to the downward motion.

- (i) The equation of motion for the jumper in the first stage of the fall is

$$\ddot{x} = g - rv$$

where g is the acceleration due to gravity, r is a positive constant, and v is the velocity of the jumper.

- (1) Given that $x = 0$ and $v = 0$ initially, show that

3

$$x = \frac{g}{r^2} \ln \left(\frac{g}{g - rv} \right) - \frac{v}{r}.$$

- (2) Given that $g = 9.8 \text{ m s}^{-2}$ and $r = 0.2 \text{ s}^{-1}$, find the length, L , of the cord such that the jumper's velocity is 30 m s^{-1} when $x = L$. Give your answer to two significant figures.

1

- (ii) In the second stage of the fall, where $x > L$, the displacement x is given by

4

$$x = e^{-\frac{t}{10}} (29 \sin t - 10 \cos t) + 92$$

where t is the time in seconds after the jumper's feet pass $x = L$.

Determine whether or not the jumper's head stays out of the water.

Question 7 continues on page 13

Question 7 (continued)

(b) Let $z = \cos \theta + i \sin \theta$.

(i) Show that $z^n + z^{-n} = 2 \cos n\theta$, where n is a positive integer. **2**

(ii) Let m be a positive integer. Show that **3**

$$(2 \cos \theta)^{2m} = 2 \left[\cos 2m\theta + \binom{2m}{1} \cos(2m-2)\theta + \binom{2m}{2} \cos(2m-4)\theta \right. \\ \left. + \cdots + \binom{2m}{m-1} \cos 2\theta \right] + \binom{2m}{m}.$$

(iii) Hence, or otherwise, prove that **2**

$$\int_0^{\frac{\pi}{2}} \cos^{2m} \theta \, d\theta = \frac{\pi}{2^{2m+1}} \binom{2m}{m}$$

where m is a positive integer.

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that **2**

$$\cot \theta + \frac{1}{2} \tan \frac{\theta}{2} = \frac{1}{2} \cot \frac{\theta}{2}.$$

- (ii) Use mathematical induction to prove that, for integers $n \geq 1$, **3**

$$\sum_{r=1}^n \frac{1}{2^{r-1}} \tan \frac{x}{2^r} = \frac{1}{2^{n-1}} \cot \frac{x}{2^n} - 2 \cot x.$$

- (iii) Show that **2**

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{2^{r-1}} \tan \frac{x}{2^r} = \frac{2}{x} - 2 \cot x.$$

- (iv) Hence find the exact value of **2**

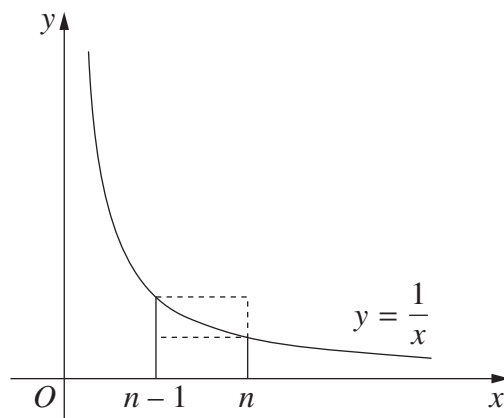
$$\tan \frac{\pi}{4} + \frac{1}{2} \tan \frac{\pi}{8} + \frac{1}{4} \tan \frac{\pi}{16} + \dots$$

Question 8 continues on page 15

Question 8 (continued)

(b)

2



Let n be a positive integer greater than 1.

The area of the region under the curve $y = \frac{1}{x}$ from $x = n - 1$ to $x = n$ is between the areas of two rectangles, as shown in the diagram.

Show that

$$e^{-\frac{n}{n-1}} < \left(1 - \frac{1}{n}\right)^n < e^{-1}.$$

- (c) A game is being played by n people, $A_1, A_2, \dots, A_n$, sitting around a table. Each person has a card with their own name on it and all the cards are placed in a box in the middle of the table. Each person in turn, starting with A_1 , draws a card at random from the box. If the person draws their own card, they win the game and the game ends. Otherwise, the card is returned to the box and the next person draws a card at random. The game continues until someone wins.

Let W be the probability that A_1 wins the game.

Let $p = \frac{1}{n}$ and $q = 1 - \frac{1}{n}$.

- (i) Show that $W = p + q^n W$.

1

- (ii) Let m be a fixed positive integer and let W_m be the probability that A_1 wins in no more than m attempts.

3

Use part (b) to show that, if n is large, $\frac{W_m}{W}$ is approximately equal to $1 - e^{-m}$.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2009 HSC Mathematics Extension 2 Marking Guidelines

Question 1 (a)

Outcomes assessed: E8

MARKING GUIDELINES

Criteria	Marks
• Correct primitive	2
• Attempts substitution, or equivalent merit	1

Question 1 (b)

Outcomes assessed: E8

MARKING GUIDELINES

Criteria	Marks
• Correct primitive	2
• Attempts integration by parts	1

Question 1 (c)

Outcomes assessed: E8

MARKING GUIDELINES

Criteria	Marks
• Correct primitive	3
• Rearranges correctly and attempts to apply appropriate standard integral	2
• Rearranges the integral into an appropriate form, or equivalent merit	1

Question 1 (d)*Outcomes assessed: E8***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	4
• Finds correct primitive, or equivalent merit	3
• Obtains correct partial fraction decomposition, or equivalent progress	2
• Attempts partial fraction decomposition, or equivalent merit	1

Question 1 (e)*Outcomes assessed: E8***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	4
• Makes substantial progress	3
• Correctly substitutes $x = \tan \theta$, or equivalent merit	2
• Attempts an appropriate substitution	1

Question 2 (a)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (b)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (c) (i)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (c) (ii)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (c) (iii)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 2 (d)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct region	2
• Sketches one of the regions	1

Question 2 (e) (i)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Writes -1 in modulus-argument form, or equivalent merit	1

Question 2 (e) (ii)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct sketch	1

Question 2 (f) (i)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	3
• Writes $a^2 - b^2 = 3$ and $2ab = 4$, or equivalent merit	2
• Writes $(a + ib)^2 = 3 + 4i$, or equivalent merit	1

Question 2 (f) (ii)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct solutions	2
• Obtains $z = \frac{x - i \pm \sqrt{3 + 4i}}{2}$, or equivalent merit	1

Question 3 (a) (i)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct sketch	2
• Indicates some important features of the graph	1

Question 3 (a) (ii)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct sketch	2
• Indicates some important features of the graph	1

Question 3 (a) (iii)

Outcomes assessed: E2, E6

MARKING GUIDELINES

Criteria	Marks
• Correct sketch	2
• Indicates some important features of the graph	1

Question 3 (b)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Shows some understanding of implicit differentiation	1

Question 3 (c)*Outcomes assessed: E4***MARKING GUIDELINES**

Criteria	Marks
• Finds correct values for a and b	3
• Finds two equations in a and b , or equivalent merit	2
• Evaluates $P(1)$ to show $6 + a + b = 0$, or equivalent merit	1

Question 3 (d)*Outcomes assessed: E7***MARKING GUIDELINES**

Criteria	Marks
• Correct answer	3
• Makes substantial progress	2
• Shows some understanding of the method of cylindrical shells	1

Question 4 (a) (i)

Outcomes assessed: E3, E6

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Differentiates correctly to find the gradient of the tangent, or equivalent merit	1

Question 4 (a) (ii)

Outcomes assessed: E4, E6

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Correctly finds the coordinates of N in terms of a and b , or equivalent merit	1

Question 4 (a) (iii)

Outcomes assessed: E3, E4, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 4 (a) (iv)

Outcomes assessed: PE2, E2, E4, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 4 (b) (i)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Correctly resolves the forces in one direction	1

Question 4 (b) (ii)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Correctly derives T , or finds N	1

Question 4 (b) (iii)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 4 (b) (iv)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (a) (i)

Outcomes assessed: PE3, E2, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Makes one relevant observation, with justification	1

Question 5 (a) (ii)

Outcomes assessed: PE3, E2, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Makes one relevant observation, with justification	1

Question 5 (a) (iii)

Outcomes assessed: PE3, E2, E9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes one relevant observation, with justification	1

Question 5 (b) (i)

Outcomes assessed: E8

MARKING GUIDELINES

Criteria	Marks
• Correct derivation	2
• Shows some understanding of integration by parts, or equivalent merit	1

Question 5 (b) (ii)

Outcomes assessed: E8

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Makes substantial progress	1

Question 5 (c) (i)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Finds $f''(x)$ correctly	1

Question 5 (c) (ii)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Attempts to use the result from part (i) , or equivalent merit	1

Question 5 (c) (iii)

Outcomes assessed: E6

MARKING GUIDELINES

Criteria	Marks
• Correct proof	1

Question 6 (a)

Outcomes assessed: E7

MARKING GUIDELINES

Criteria	Marks
• Correct answer	3
• Makes substantial progress	2
• Finds the correct expression for cross-sectional area, or equivalent merit	1

Question 6 (b) (i)

Outcomes assessed: E4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 6 (b) (ii) (1)

Outcomes assessed: E3, E4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 6 (b) (ii) (2)

Outcomes assessed: E3, E4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 6 (c) (i)

Outcomes assessed: PE6, E4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 6 (c) (ii)

Outcomes assessed: PE6, E4

MARKING GUIDELINES

Criteria	Marks
• Derives equation correctly	2
• Writes $PQ = c - x$	1

Question 6 (c) (iii)

Outcomes assessed: PE6, E4

MARKING GUIDELINES

Criteria	Marks
• Finds S correctly	2
• Finds the vertex, or the focal length, or equivalent merit	1

Question 6 (c) (iv)

Outcomes assessed: PE6, E4, E9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes some progress towards a correct solution	1

Question 7 (a) (i) (1)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct solution, including consideration of initial condition	3
• Correctly integrates with respect to v • Correctly makes $\ddot{x}$ the subject, or equivalent merit	2
• Write $\frac{dx}{dv} = \frac{V}{g - rv}$ OR • Differentiates x with respect to t correctly, or equivalent merit	1

Question 7 (a) (i) (2)

Outcomes assessed: E5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 7 (a) (ii)

Outcomes assessed: E2, E5, E9

MARKING GUIDELINES

Criteria	Marks
• Correct answer	4
• Finds x correctly, or equivalent merit	3
• Finds t correctly, or equivalent merit	2
• Finds $\frac{dx}{dt}$ correctly, or equivalent merit	1

Question 7 (b) (i)

Outcomes assessed: E3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Shows knowledge of de Moivre's theorem	1

Question 7 (b) (ii)

Outcomes assessed: HE3, E2, E4, E9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Applies binomial theorem, or equivalent merit	2
• Recognises link to $(\bar{i})$	1

Question 7 (b) (iii)

Outcomes assessed: E2, E8, E9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Recognises how to use the result in part (ii) , or equivalent merit	1

Question 8 (a) (i)

Outcomes assessed: PE3, E2, E4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 8 (a) (ii)

Outcomes assessed: HE2, E2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Establishes the induction step	2
• Verifies the result for $n = 1$	1

Question 8 (a) (iii)

Outcomes assessed: HE7, E2, E4, E9

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 8 (a) (iv)

Outcomes assessed: HE7, E4

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Substitutes $x = \frac{\pi}{2}$ in the result in part (iii) , or equivalent merit	1

Question 8 (b)

Outcomes assessed: H8, HE7, E2, E6, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	2
• Obtains $\frac{1}{n-1} < [\ln x]_{n-1}^n < \frac{1}{n}$, or equivalent merit	1

Question 8 (c) (i)

Outcomes assessed: HE3, E2, E9

MARKING GUIDELINES

Criteria	Marks
• Correct explanation	1

Question 8 (c) (ii)

Outcomes assessed: HE3, E2, E6, E9

MARKING GUIDELINES

Criteria	Marks
• Correct proof	3
• Finds an expression for $\frac{W_m}{W}$ and attempts to use part (b), or equivalent merit	2
• Finds an expression for W_m , or equivalent merit	1

Mathematics Extension 2

2009 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	2	4.1	E8
1 (b)	2	4.1	E8
1 (c)	3	4.1	E8
1 (d)	4	4.1	E8
1 (e)	4	4.1	E8
2 (a)	1	2.1	E3
2 (b)	1	2.1	E3
2 (c) (i)	1	2.2	E3
2 (c) (ii)	1	2.2	E3
2 (c) (iii)	1	2.2	E3
2 (d)	2	2.5	E3
2 (e) (i)	2	2.4	E3
2 (e) (ii)	1	2.2, 2.4	E3
2 (f) (i)	3	2.1	E3
2 (f) (ii)	2	2.1	E3
3 (a) (i)	2	1.5	E6
3 (a) (ii)	2	1.6	E6
3 (a) (iii)	2	1.8	E2, E6
3 (b)	3	1.8	E6
3 (c)	3	7.2	E4
3 (d)	3	5.1	E7
4 (a) (i)	2	3.1	E3, E6
4 (a) (ii)	2	3.1	E4, E6
4 (a) (iii)	2	3.1	E3, E4, E9
4 (a) (iv)	2	3.1	PE2, E2, E4, E9
4 (b) (i)	2	6.3.3	E5
4 (b) (ii)	2	6.3.3	E5
4 (b) (iii)	2	6.3.3	E5
4 (b) (iv)	1	6.3.3	E5
5 (a) (i)	2	8.1	PE3, E2, E9

Question	Marks	Content	Syllabus outcomes
5 (a) (ii)	2	8.1	PE3, E2, E9
5 (a) (iii)	2	8.1	PE3, E2, E9
5 (b) (i)	2	4.1	E8
5 (b) (ii)	2	4.1	E8
5 (c) (i)	2	8.0	E6
5 (c) (ii)	2	8.0	E6
5 (c) (iii)	1	8.0	E6
6 (a)	3	5.1	E7
6 (b) (i)	1	7.4	E4
6 (b) (ii) (1)	2	2.2, 7.4	E3, E4
6 (b) (ii) (2)	2	2.1, 7.4, 7.5	E3, E4
6 (c) (i)	1	8.0	PE6, E4
6 (c) (ii)	2	8.0	PE6, E4
6 (c) (iii)	2	8.0	PE6, E4
6 (c) (iv)	2	8.0	PE6, E4, E9
7 (a) (i) (1)	3	6.2.3	E5
7 (a) (i) (2)	1	6.2.3	E5
7 (a) (ii)	4	6.2.3	E2, E5, E9
7 (b) (i)	2	2.4	E3
7 (b) (ii)	3	8.0	HE3, E2, E4, E9
7 (b) (iii)	2	4.1	E2, E8, E9
8 (a) (i)	2	8.0	PE3, E2, E4
8 (a) (ii)	3	8.2	HE2, E2
8 (a) (iii)	2	8.0	HE7, E2, E4, E9
8 (a) (iv)	2	8.0	HE7, E4
8 (b)	2	8.0	H8, HE7, E2, E6, E9
8 (c) (i)	1	8.0	HE3, E2, E9
8 (c) (ii)	3	8.0	HE3, E2, E6, E9